

# Lecture 8b: Non-linear Change

March 13, 2025

# From Last Lecture...

- Time varying-covariates
  - Main effect of time-varying covariates
  - The effect of time-varying covariates on the slope
- Loglikelihood Ratio Test

# Learning Objectives

- Variably spaced measurement occasions
  - Selecting a time variable
- Varying number of measurement occasions
  - Differing number of waves per person
  - Other issues to consider
    - Boundary constraints
    - Nonconvergence
    - Missing data concerns
- Discontinuous individual change
  - Types of discontinuous level-1 models for change
  - Selecting the best model of discontinuous level-1 change
  - Extensions of discontinuous growth models

# Variably Spaced Measurement Occasions

# The thing about time....

(Singer and Willet pgs. 138-140)

- Two key features of longitudinal data—
  - **Balanced:** Everyone in the sample has the same number of timepoints
    - If this is not true, we call these data **unbalanced**
    - Researchers try to collect balanced data 😊
  - **Time-structured:** Timepoints are consistent for all individuals in the sample
- How do irregularities arise?
  - The realities of data collection
    - Ginexi et al. (e.g. time from unemployment)
    - Missing data
    - Funding delays/gaps – Flint Adolescent Study (PI: Zimmerman)
- Can also build in timing irregularities
  - Accelerated cohort studies
    - Many single cohorts where individuals start at different ages

# Variable selection in variably spaced datasets

(Singer and Willet pgs. 140-144)

- Time can be treated in many different ways
  - Must decide which timing variable makes the most sense theoretically/conceptually/logically
    - Wave, age, time-to-event/time-from-event
  - Must decide which timing variable provides the best model fit
    - Does the model fit the data well?
    - What do the AIC/BIC suggest?

# Variable selection in variably spaced datasets

# *Example for handling variably spaced waves: Reading achievement over time*

**Data source:** *Children of the National Longitudinal Survey of Youth (CNLSY)*

## Sample: 89 children

-  *Each approximately 6 years old at study start*

## Research design

-  *3 waves of data collected in 1986, 1988, and 1990, when the children were to be “in their 6th yr,” “in their 8th yr,” and “in their 10th yr”*
-  *Of course, not each child was tested on his/her birthday or half-birthday, which creates the variably spaced waves*
-  *The outcome, PIAT, is the child’s unstandardized score on the reading portion of the Peabody Individual Achievement Test*
  -  *Not standardized for age so we can see growth over time*
-  *No substantive predictors to keep the example simple*

## Research question

-  *How do PIAT scores change over time?*



# Variable selection in variably spaced datasets

- **Wave:** Design variable, not very informative on its own
- **AGEGRP:** Expected age on the day data collected (time structured)
- **AGE:** The actual age of child on the day data collected (time varies by person, unstructured)
- Cadence of the data does not matter we can still model change either way
  - More important functional form of the data

Table 5.1: Excerpts from the person-period data set for the reading study

| ID  | WAVE | AGEGRP | AGE   | PIAT |
|-----|------|--------|-------|------|
| 04  | 1    | 6.5    | 6.00  | 18   |
| 04  | 2    | 8.5    | 8.50  | 31   |
| 04  | 3    | 10.5   | 10.67 | 50   |
| 27  | 1    | 6.5    | 6.25  | 19   |
| 27  | 2    | 8.5    | 9.17  | 36   |
| 27  | 3    | 10.5   | 10.92 | 57   |
| 31  | 1    | 6.5    | 6.33  | 18   |
| 31  | 2    | 8.5    | 8.83  | 31   |
| 31  | 3    | 10.5   | 10.92 | 51   |
| 33  | 1    | 6.5    | 6.33  | 18   |
| 33  | 2    | 8.5    | 8.92  | 34   |
| 33  | 3    | 10.5   | 10.75 | 29   |
| 41  | 1    | 6.5    | 6.33  | 18   |
| 41  | 2    | 8.5    | 8.75  | 28   |
| 41  | 3    | 10.5   | 10.83 | 36   |
| 49  | 1    | 6.5    | 6.50  | 19   |
| 49  | 2    | 8.5    | 8.75  | 32   |
| 49  | 3    | 10.5   | 10.67 | 48   |
| 69  | 1    | 6.5    | 6.67  | 26   |
| 69  | 2    | 8.5    | 9.17  | 47   |
| 69  | 3    | 10.5   | 11.33 | 45   |
| 77  | 1    | 6.5    | 6.83  | 17   |
| 77  | 2    | 8.5    | 8.08  | 19   |
| 77  | 3    | 10.5   | 10.00 | 28   |
| 87  | 1    | 6.5    | 6.92  | 22   |
| 87  | 2    | 8.5    | 9.42  | 49   |
| 87  | 3    | 10.5   | 11.50 | 64   |
| ... | ...  | ...    | ...   | ...  |

Note that *TIME* is clocked using three distinct variables: *WAVE*, *AGEGRP*, and *AGE*.

# Variable selection

$$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$$

$$\pi_{0i} = \gamma_{00} + \zeta_{0i}$$

$$\pi_{1i} = \gamma_{10} + \zeta_{1i},$$

where

$$\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2) \text{ and } \begin{bmatrix} \zeta_{0i} \\ \zeta_{1i} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_0^2 & \sigma_{01} \\ \sigma_{10} & \sigma_1^2 \end{bmatrix}\right)$$

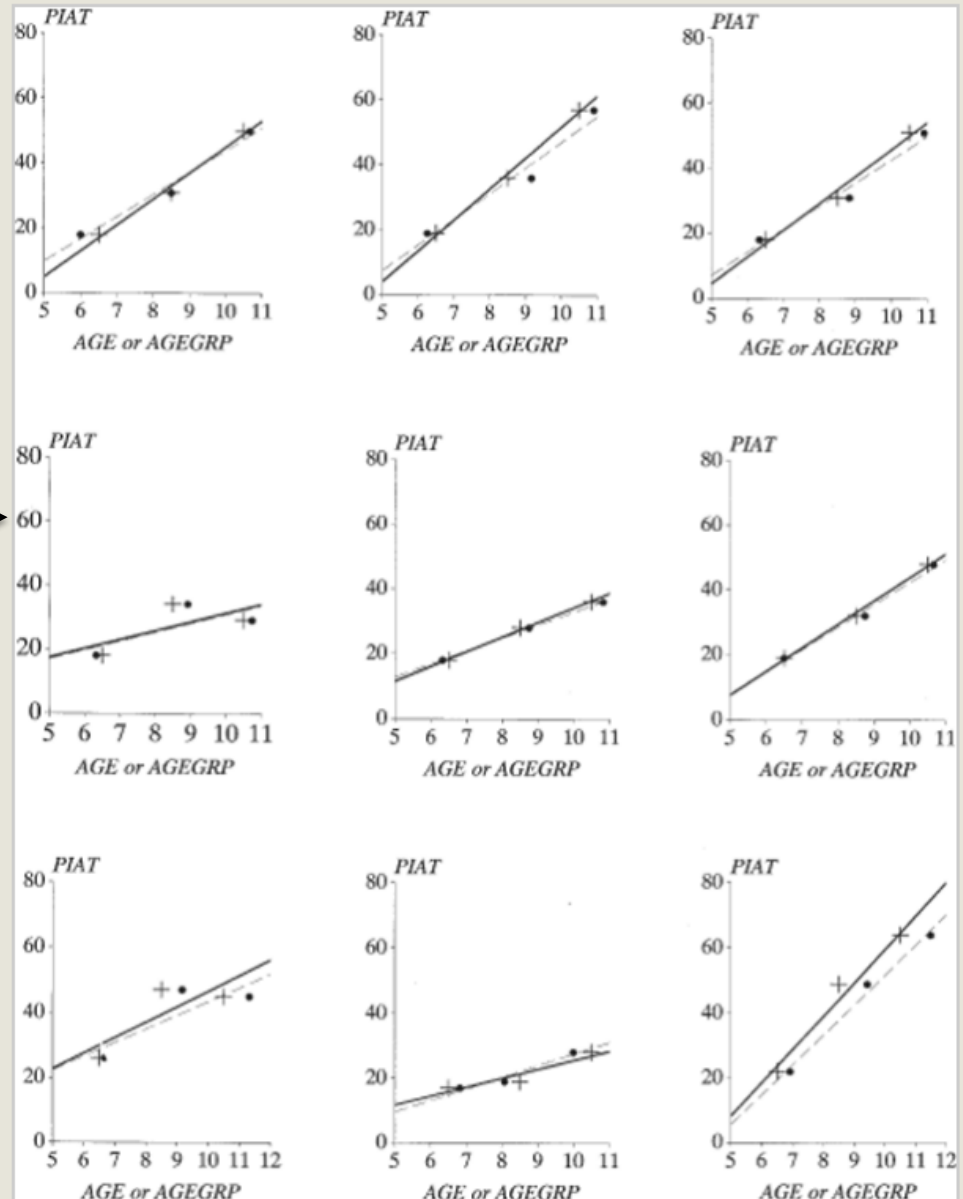
(5.1b)

- Comparing time-structured and time-unstructured representations of the effect of *TIME*.
- Empirical change plots with superimposed OLS trajectories for 9 participants in the reading study.
- The “+’s” and solid lines are for *TIME* clocked using the child’s *target* age at data collection; the •’s and dashed lines are for *TIME* clocked using each child’s observed age

**The steps of model evaluation are the same from here**

1. What is our question?
2. Fit unconditional model using both time variables
3. Assess fit

**Does it appear that there is much of a difference between the the two growth models using the two different age variables?**



# Variable selection

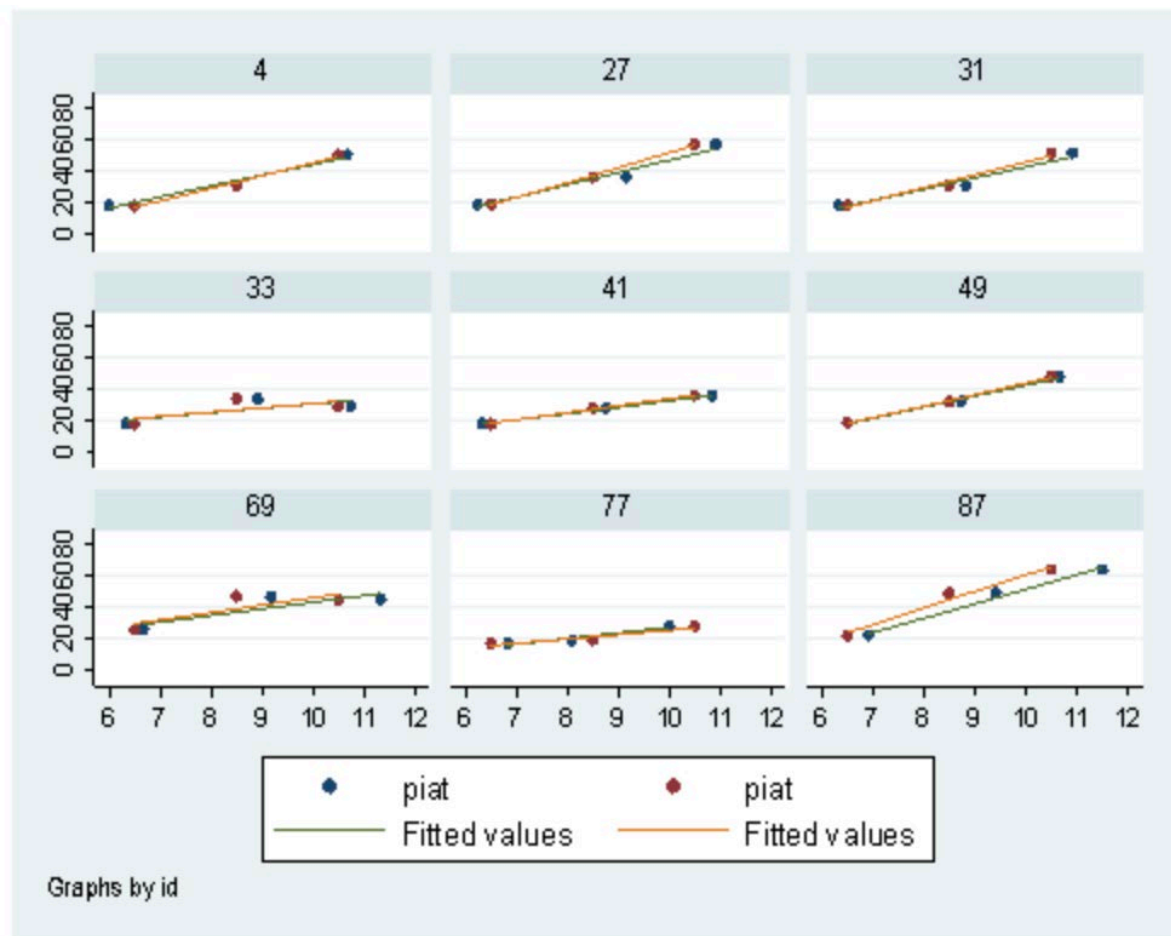
- Comparing time-structured and time-unstructured representations of the effect of *TIME*.
- Empirical change plots with superimposed OLS trajectories for 9 participants in the reading study.
- The “+’s” and solid lines are for *TIME* clocked using the child’s *target* age at data collection; the •’s and dashed lines are for *TIME* clocked using each child’s observed age

## The steps of model evaluation are the same from here

1. What is our question?
2. Fit unconditional model using both time variables
3. Assess fit

## Does it appear that there is much of a difference between the the two growth models using the two different age variables?

- Not based on the OLS trajectory plot



# Variable selection

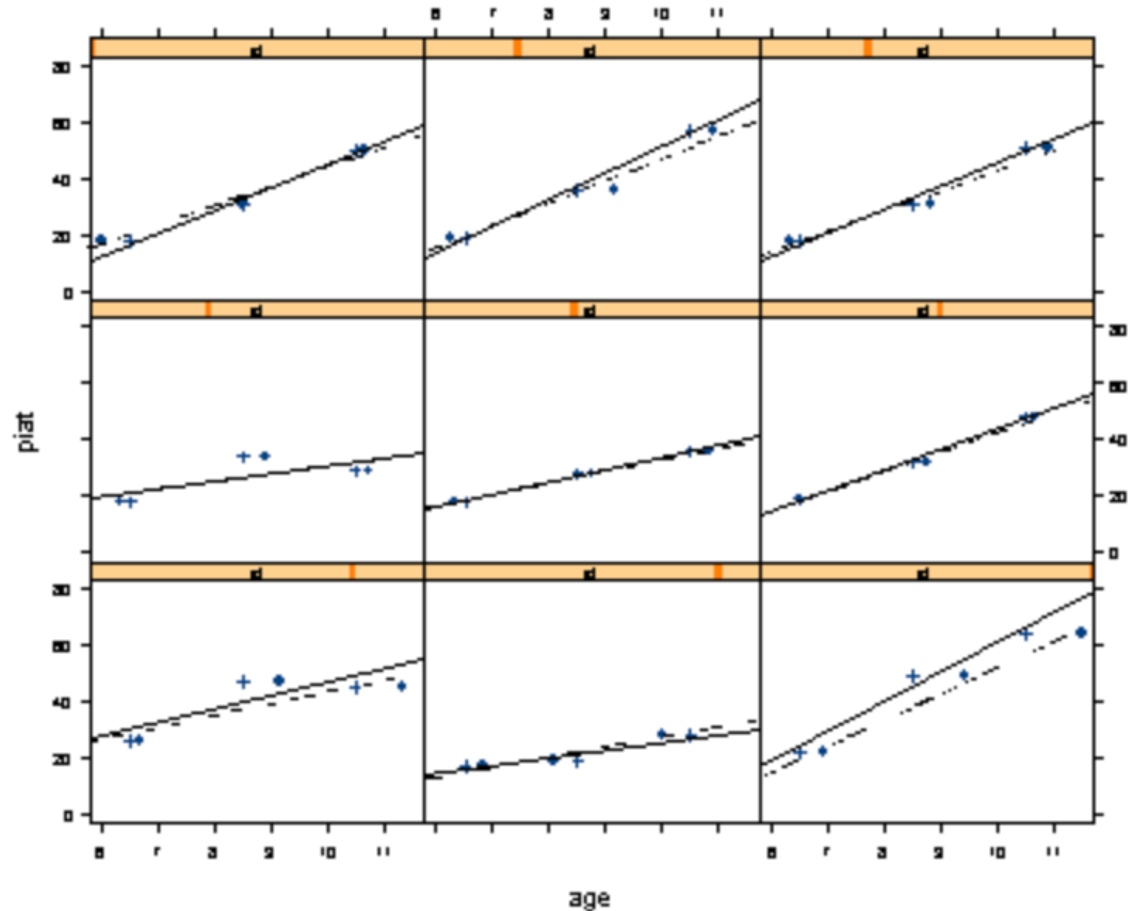
- Comparing time-structured and time-unstructured representations of the effect of *TIME*.
- Empirical change plots with superimposed OLS trajectories for 9 participants in the reading study.
- The “+’s” and solid lines are for *TIME* clocked using the child’s *target* age at data collection; the •’s and dashed lines are for *TIME* clocked using each child’s observed age

**The steps of model evaluation are the same from here**

1. What is our question?
2. Fit unconditional model using both time variables
3. Assess fit

**Does it appear that there is much of a difference between the the two growth models using the two different age variables?**

- Not based on the OLS trajectory plot



# Variable selection in variably spaced datasets

**Table 5.2: Results of using alternative representations for the main effect of *TIME* ( $n = 89$ ) when fitting an unconditional growth model to the CNLSY reading data**

|                                                                             |                             |                   | Predictor representing TIME |            |
|-----------------------------------------------------------------------------|-----------------------------|-------------------|-----------------------------|------------|
|                                                                             |                             |                   | AGEGRP – 6.5                | AGE – 6.5  |
|                                                                             |                             |                   | Parameter                   |            |
| Fixed Effects                                                               |                             |                   |                             |            |
| Initial reading score at 6.5 yrs.                                           | Initial status, $\eta_{0i}$ | Intercept         | $Y_{00}$                    | 21.1629*** |
|                                                                             |                             |                   |                             | (0.6143)   |
| Avg. rate of change in reading score from 6-11 yrs.                         | Rate of Change, $\eta_{1i}$ | Intercept         | $Y_{10}$                    | 5.0309***  |
|                                                                             |                             |                   |                             | (0.2956)   |
| Variance Components                                                         |                             |                   |                             |            |
| Within child scatter of reading scores around his/her own true change score | Level-1:                    | within-person     | $\sigma_{\epsilon}^2$       | 27.04***   |
|                                                                             |                             |                   |                             | 27.45***   |
| Between child variability in the intercept and slope                        | Level-2:                    | In initial status | $\sigma_0^2$                | 11.05*     |
|                                                                             |                             | In rate of change | $\sigma_1^2$                | 4.40***    |
| Goodness-of-fit                                                             |                             |                   |                             |            |
|                                                                             |                             |                   | Deviance                    | 1819.8     |
|                                                                             |                             |                   | AIC                         | 1831.9     |
|                                                                             |                             |                   | BIC                         | 1846.9     |

~  $p < .10$ ; \*  $p < .05$ ; \*\*  $p < .01$ ; \*\*\*  $p < .001$ .

# Variable selection in variably spaced datasets

**Table 5.2: Results of using alternative representations for the main effect of *TIME* ( $n = 89$ ) when fitting an unconditional growth model to the CNLSY reading data**

|                             |                   |                       | Predictor representing TIME |                        |
|-----------------------------|-------------------|-----------------------|-----------------------------|------------------------|
|                             |                   |                       | AGEGRP – 6.5                | AGE – 6.5              |
| Fixed Effects               |                   |                       |                             |                        |
| Initial status, $\eta_{0i}$ | Intercept         | $Y_{00}$              | 21.1629***<br>(0.6143)      | 21.0608***<br>(0.5593) |
| Rate of Change, $\eta_{1i}$ | Intercept         | $Y_{10}$              | 5.0309***<br>(0.2956)       | 4.5400***<br>(0.2606)  |
| Variance Components         |                   |                       |                             |                        |
| Level-1:                    | within-person     | $\sigma_{\epsilon}^2$ | 27.04***                    | 27.45***               |
| Level-2:                    | In initial status | $\sigma_0^2$          | 11.05*                      | 5.11                   |
|                             | In rate of change | $\sigma_1^2$          | 4.40***                     | 3.30***                |
| Goodness-of-fit             |                   |                       |                             |                        |
|                             | Deviance          |                       | 1819.8                      | 1803.9                 |
|                             | AIC               |                       | 1831.9                      | 1815.9                 |
|                             | BIC               |                       | 1846.9                      | 1830.8                 |

~  $p < .10$ ; \*  $p < .05$ ; \*\*  $p < .01$ ; \*\*\*  $p < .001$ .

**Treating this unstructured data set as though it is time-structured introduces error into the analysis!**

Very little difference between models in intercept

Rate of change half point larger for AGEGRP  
= 2 point difference in PIAT over 4 years

Very little difference between models in level-1 variance

Level-2 residuals larger for model with AGEGRP compared to model with AGE

- **AGEGRP** has regular intervals which are shorter in duration. Thus, a steeper slope
- **AGEGRP** has regular large level-2 variances because it fits less well

# Variable selection in variably spaced datasets

**Table 5.2: Results of using alternative representations for the main effect of *TIME* ( $n = 89$ ) when fitting an unconditional growth model to the CNLSY reading data**

|                             |                   |                       | Predictor representing <i>TIME</i> |                        |
|-----------------------------|-------------------|-----------------------|------------------------------------|------------------------|
|                             |                   |                       | AGEGRP – 6.5                       | AGE – 6.5              |
| Fixed Effects               |                   |                       |                                    |                        |
| Initial status, $\eta_{0i}$ | Intercept         | $\gamma_{00}$         | 21.1629***<br>(0.6143)             | 21.0608***<br>(0.5593) |
| Rate of Change, $\eta_{1i}$ | Intercept         | $\gamma_{10}$         | 5.0309***<br>(0.2956)              | 4.5400***<br>(0.2606)  |
| Variance Components         |                   |                       |                                    |                        |
| Level-1:                    | within-person     | $\sigma_{\epsilon}^2$ | 27.04***                           | 27.45***               |
| Level-2:                    | In initial status | $\sigma_0^2$          | 11.05*                             | 5.11                   |
|                             | In rate of change | $\sigma_1^2$          | 4.40***                            | 3.30***                |
| Goodness-of-fit             |                   |                       |                                    |                        |
|                             | Deviance          |                       | 1819.8                             | 1803.9                 |
|                             | AIC               |                       | 1831.9                             | 1815.9                 |
|                             | BIC               |                       | 1846.9                             | 1830.8                 |

~  $p < .10$ ; \*  $p < .05$ ; \*\*  $p < .01$ ; \*\*\*  $p < .001$ .

- AIC/BIC better for model with AGE compared to model with AGEGRP
- Recall that AIC/BIC can be compared across non-nested models while Deviance cannot

Varying Number of Timepoints

# Varying number of timepoints

(Singer and Willet pgs. 146-159)

- Measurement occasions can have varying spacing in which you may want to account for (discussed in last section)
  - Example – actual age vs. predicted age
- Individuals can also have differing numbers of measurement occasion
  - Example – 1 measurement occasion vs. 13 measurement occasions
- Multilevel modeling in comparison to other techniques (e.g. repeated measures ANOVA) can handle both of these issues as long as the data is properly specified

# Varying number of timepoints

- National Longitudinal Survey of Youth (NLSY) example
  - Attempting to understand labor market experiences of male high school dropouts
  - Natural logarithm of wages (**LWN**) in years
  - The moment to the nearest day in an individuals work history (**EXPER**)
  - Highest Grade Completed 6-12 (**HGC**); Centered at 9<sup>th</sup> grade (**HGC-9**)
  - Race/Ethnicity – Black vs. White/Latino (**BLACK**)
  - N = 888 males
- Complicated dataset...
  1. Starting age 14-17
  2. Waves of data collection have 1 and 2 year gaps
  3. Data collection conducted during different times of the year
  4. Sometimes had more than one job simultaneously
  5. Entered and exited labor force at different times

Murane et al. decided the best way to characterize time would be to track wages from each participants first day of work.

# Varying number of timepoints

- Individual 206 only has 3 timepoints
  - 1.874, 2.814, and 4.314 years of experience post labor force entry

- $e^{(2.028)} = \$7.60$  per hour
- $e^{(2.297)} = \$9.90$  per hour
- $e^{(2.482)} = \$9.50$  per hour

- Individuals have different #'s of timepoints
  - eg ID 332 has 10 timepoints
- Timing of timepoints within individual are unequally spaced
- We can include individuals with only 1 or 2 timepoints

Table 5.3: Excerpts from the person-period data set for the high school dropout wage study

| ID   | EXPER | LNW   | BLACK | HGC | UERATE |
|------|-------|-------|-------|-----|--------|
| 206  | 1.874 | 2.028 | 0     | 10  | 9.200  |
| 206  | 2.814 | 2.297 | 0     | 10  | 11.000 |
| 206  | 4.314 | 2.482 | 0     | 10  | 6.295  |
| 332  | 0.125 | 1.630 | 0     | 8   | 7.100  |
| 332  | 1.625 | 1.476 | 0     | 8   | 9.600  |
| 332  | 2.413 | 1.804 | 0     | 8   | 7.200  |
| 332  | 3.393 | 1.439 | 0     | 8   | 6.195  |
| 332  | 4.470 | 1.748 | 0     | 8   | 5.595  |
| 332  | 5.178 | 1.526 | 0     | 8   | 4.595  |
| 332  | 6.082 | 2.044 | 0     | 8   | 4.295  |
| 332  | 7.043 | 2.179 | 0     | 8   | 3.395  |
| 332  | 8.197 | 2.186 | 0     | 8   | 4.395  |
| 332  | 9.092 | 4.035 | 0     | 8   | 6.695  |
| 1028 | 0.004 | 0.872 | 1     | 8   | 9.300  |
| 1028 | 0.035 | 0.903 | 1     | 8   | 7.400  |
| 1028 | 0.515 | 1.389 | 1     | 8   | 7.300  |
| 1028 | 1.483 | 2.324 | 1     | 8   | 7.400  |
| 1028 | 2.141 | 1.484 | 1     | 8   | 6.295  |
| 1028 | 3.161 | 1.705 | 1     | 8   | 5.895  |
| 1028 | 4.103 | 2.343 | 1     | 8   | 6.900  |

# Varying number of timepoints

Table 5.4: Results of fitting a taxonomy of multilevel models for change to the high school dropout wage data ( $n = 888$ )

|                                                                |                   | Parameter                | Model A               | Model B               | Model C                |
|----------------------------------------------------------------|-------------------|--------------------------|-----------------------|-----------------------|------------------------|
| Fixed Effects                                                  |                   |                          |                       |                       |                        |
| Initial status, $\pi_{0j}$                                     | Intercept         | $Y_{00}$                 | 1.7156***<br>(0.0108) | 1.7171***<br>(0.0125) | 1.7215***<br>(0.0107)  |
|                                                                | (HGC - 9)         | $Y_{01}$                 |                       | 0.0349***<br>(0.0079) | 0.0384***<br>(0.0064)  |
|                                                                | BLACK             | $Y_{02}$                 |                       | 0.0154<br>(0.0239)    |                        |
| Rate of change, $\pi_{1j}$                                     | Intercept         | $Y_{10}$                 | 0.0457***<br>(0.0023) | 0.0493***<br>(0.0026) | 0.0489***<br>(0.0025)  |
|                                                                | (HGC - 9)         | $Y_{11}$                 |                       | 0.0013<br>(0.0017)    |                        |
|                                                                | BLACK             | $Y_{12}$                 |                       | -0.0182**<br>(0.0055) | -0.0161***<br>(0.0045) |
| Variance Components                                            |                   |                          |                       |                       |                        |
| Level-1:                                                       | within-person     | $\sigma_{\varepsilon}^2$ | 0.0951***             | 0.0952***             | 0.0952***              |
| Level-2:                                                       | In initial status | $\sigma_0^2$             | 0.0543***             | 0.0518***             | 0.0518***              |
|                                                                | In rate of change | $\sigma_1^2$             | 0.0017***             | 0.0016***             | 0.0016***              |
| Goodness-of-fit                                                |                   |                          |                       |                       |                        |
|                                                                | Deviance          |                          | 4921.4                | 4873.8                | 4874.7                 |
|                                                                | AIC               |                          | 4933.4                | 4893.8                | 4890.7                 |
|                                                                | BIC               |                          | 4962.1                | 4941.7                | 4929.0                 |
| $\sim p < .10$ ; * $p < .05$ ; ** $p < .01$ ; *** $p < .001$ . |                   |                          |                       |                       |                        |

Initial inflation-adjusted wages

Highest grade completed intercept significant but not slope. Overall, those who drop-out later earn higher wages upon entering the labor market

$(e^{.0457} - 1) = 4.7$ , meaning that the average male dropouts inflation adjusted hourly wage increases 4.7% for each year of labor participation

Slope significant but not intercept. The wages of Blacks increase less rapidly compared to Whites and Latinos

There remains significant variability around the intercept and slope

# Varying number of timepoints

Table 5.4: Results of fitting a taxonomy of multilevel models for change to the high school dropout wage data ( $n = 888$ )

|                            |                   | Parameter             | Model A               | Model B               | Model C                |
|----------------------------|-------------------|-----------------------|-----------------------|-----------------------|------------------------|
| Fixed Effects              |                   |                       |                       |                       |                        |
| Initial status, $\pi_{0j}$ | Intercept         | $Y_{00}$              | 1.7156***<br>(0.0108) | 1.7171***<br>(0.0125) | 1.7215***<br>(0.0107)  |
|                            |                   | $(HGC - 9)$           |                       | 0.0349***<br>(0.0079) | 0.0384***<br>(0.0064)  |
|                            |                   | BLACK                 |                       | 0.0154<br>(0.0239)    | NS                     |
| Rate of change, $\pi_{1j}$ | Intercept         | $Y_{10}$              | 0.0457***<br>(0.0023) | 0.0493***<br>(0.0026) | 0.0489***<br>(0.0025)  |
|                            |                   | $(HGC - 9)$           |                       | 0.0013<br>(0.0017)    | NS                     |
|                            |                   | BLACK                 |                       | -0.0182**<br>(0.0055) | -0.0161***<br>(0.0045) |
| Variance Components        |                   |                       |                       |                       |                        |
| Level-1:                   | within-person     | $\sigma_{\epsilon}^2$ | 0.0951***             | 0.0952***             | 0.0952***              |
| Level-2:                   | In initial status | $\sigma_0^2$          | 0.0543***             | 0.0518***             | 0.0518***              |
|                            | In rate of change | $\sigma_1^2$          | 0.0017***             | 0.0016***             | 0.0016***              |
| Goodness-of-fit            |                   |                       |                       |                       |                        |
|                            | Deviance          |                       | 4921.4                | 4873.8                | 4874.7                 |
|                            | AIC               |                       | 4933.4                | 4893.8                | 4890.7                 |
|                            | BIC               |                       | 4962.1                | 4941.7                | 4929.0                 |

$\sim p < .10$ ; \*  $p < .05$ ; \*\*  $p < .01$ ; \*\*\*  $p < .001$ .

Initial inflation-adjusted wages

Highest grade completed intercept significant but not slope. Overall, those who drop-out later earn higher wages upon entering the labor market

$(e^{.0457} - 1) = 4.7$ , meaning that the average male dropouts inflation adjusted hourly wage increases 4.7% for each year of labor participation

Slope significant but not intercept. The wages of Blacks increase less rapidly compared to Whites and Latinos

There remains significant variability around the intercept and slope

# Varying number of timepoints

## Model C Interpretation

- Controlling for highest grade completed, the average annual percentage increase in wages is  $(e^{(.0489)} - 1) = 5\%$  for White and Latinos in comparison to  $(e^{(.0328)} - 1) = 3.3\%$  for Blacks (see section 4.5.3 for calculation)
- Race is more important than high school drop out timing
  - After about 7 years in the labor market, the effect of dropping out of college later is wiped out and Black males earn a lower hourly wage compared to White and Latino males

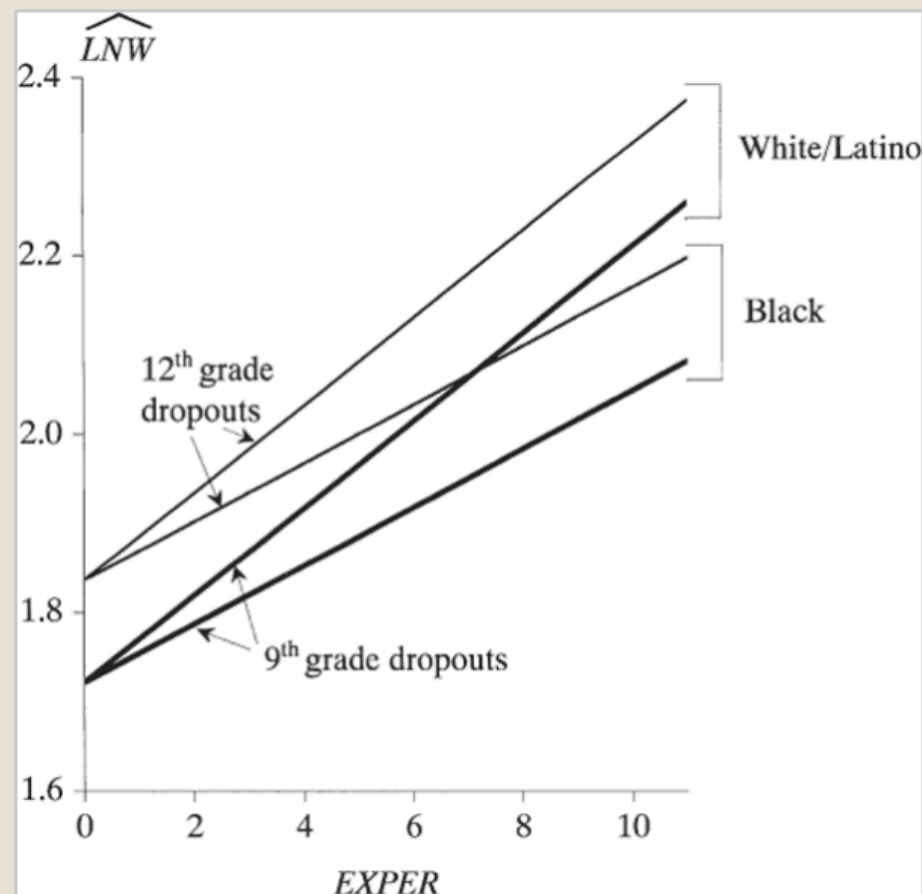
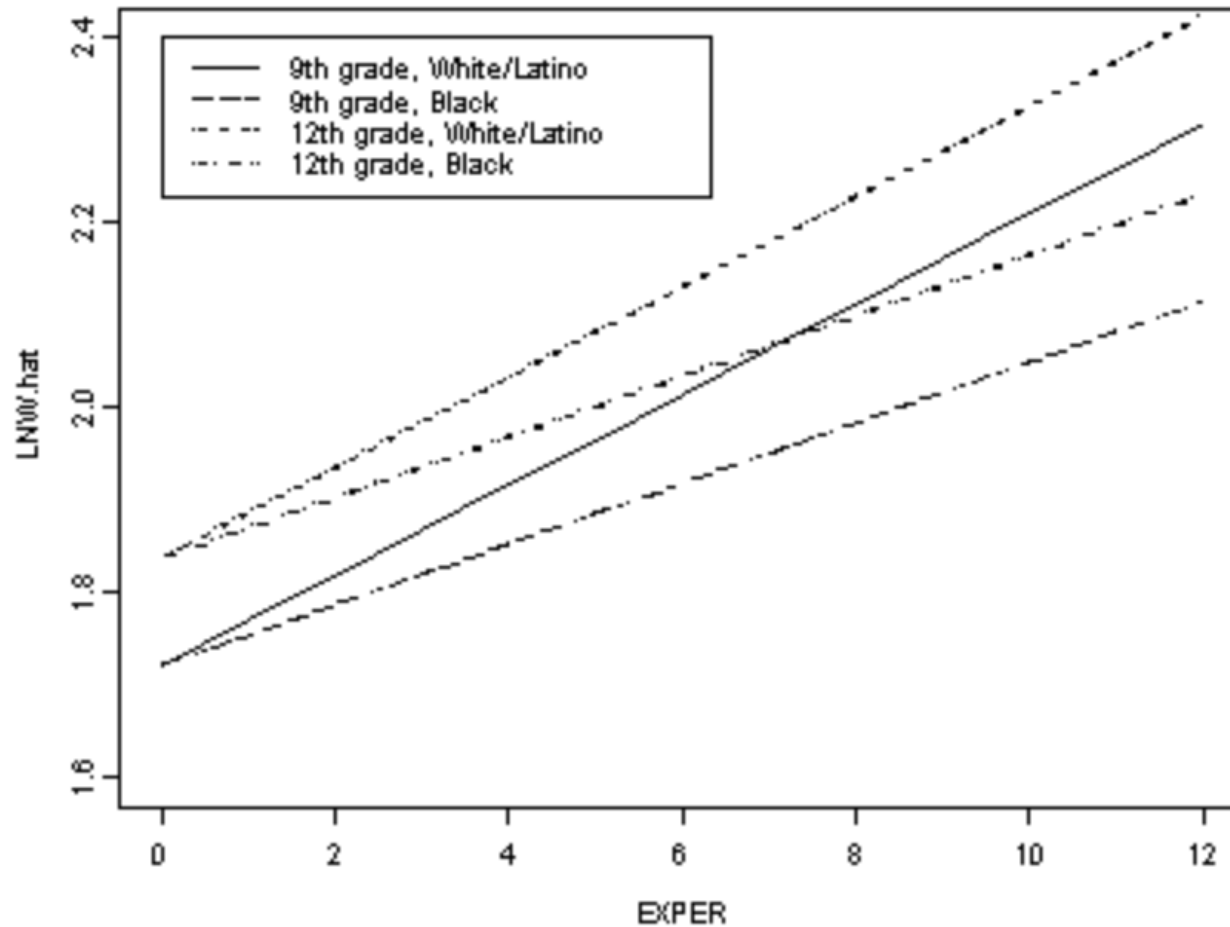
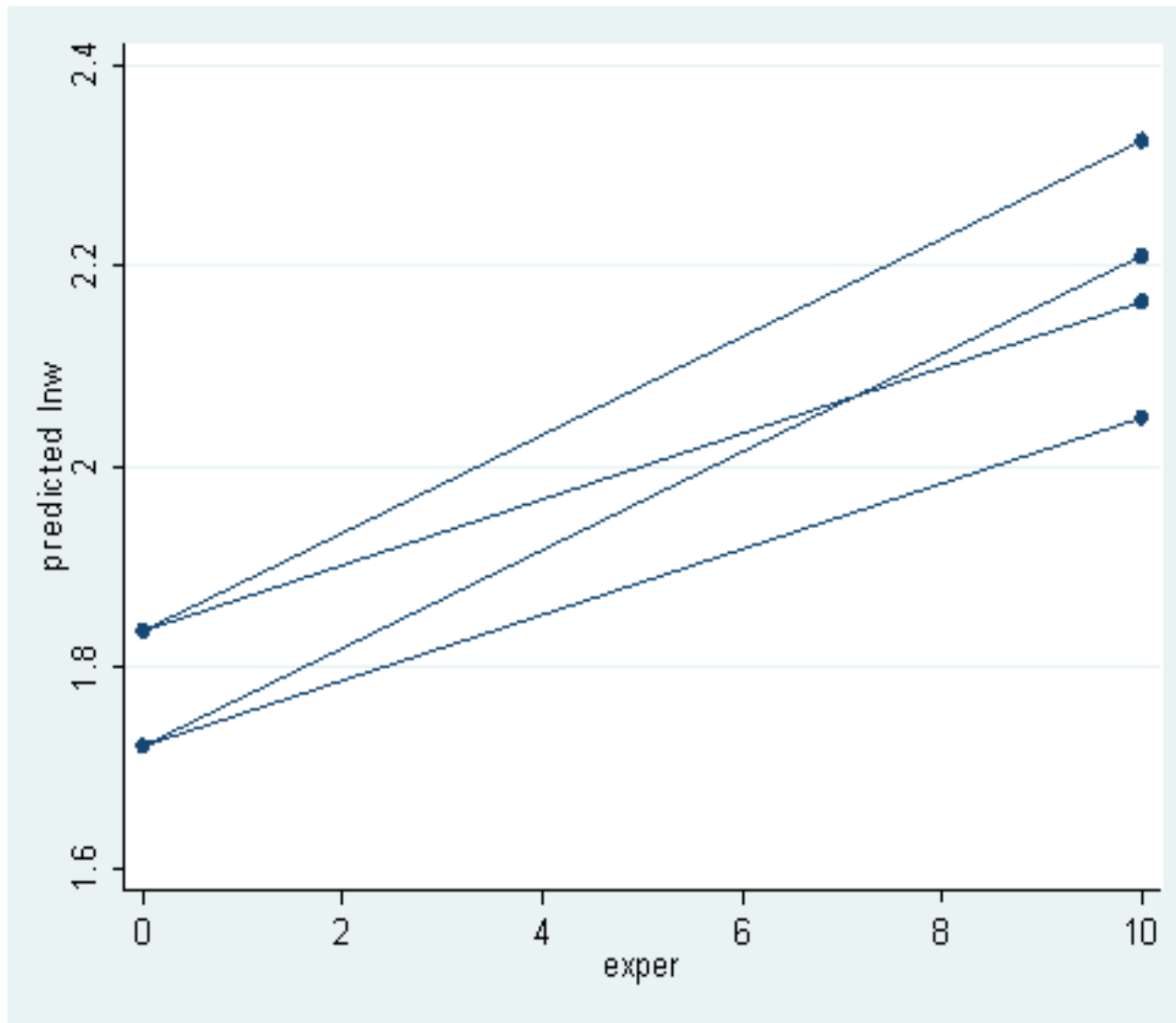


Figure 5.2. Displaying the results of a fitted multilevel model for change. Log wage trajectories from Model C of table 5.4 for four prototypical dropouts: Blacks and Whites/Latinos who dropped out in 9th and 12th grades.

# Varying number of timepoints



# Varying number of timepoints



# Other issues to consider

(Singer and Willet pgs. 152-159)

## Boundary constraints

- Data is outside of normal ranges, for instance if your covariance matrix calculates negative values
  - “Hessian is not negative semidefinite” error message in stata (see lecture 7; class exercise 1)
  - **Potential Fix:** simplify your model

## Nonconvergence

- Model is too complex
  - Highly unbalanced data
- Poorly specified models
- Insufficient data
- **Potential Fix:** increase iterations

## Missing data concerns

- Missing completely at random (MCAR) - Preferred
- Missing at random (MAR) – Can continue to model under certain conditions
- Missing not at random (MNAR) – Need a different technique

# Summary

- Multilevel modeling techniques can usually handle data with uneven timing between measurement occasions
  - Follow the steps to select the best timing variable based on concept/theory and fit statistics (unconditional growth models)
- Multilevel modelling techniques can also handle unbalanced data
  - All data included in analysis
  - May have to do some transformations or recoding to help with interpretation of the data
- There are a few issues to think through
  - Boundary constraints, nonconvergence, and missing data
    - Usually better specifying your model, increasing iterations, and/or simplifying your model will help address these issues (but not always)

## *Practical advice: Problems can arise when analyzing unbalanced data sets*



*The multilevel model for change is designed to handle unbalanced data sets, and in most circumstances, it does its job well, however...*

☛ *When imbalance is severe, or lots of people have just 1 or 2 waves of data, problems can occur*

- ☛ *You may not estimate some parameters (well)*
- ☛ *Iterative fitting algorithms may not converge*
- ☛ *Some estimates may hit boundary constraints*
- ☛ *Problem is usually manifested via VCs not fixed effects (because the fixed portion of the model is like a 'regular regression model').*

☛ *Software packages may not issue clear warning signs*

- ☛ *If you're lucky, you'll get negative variance components*
- ☛ *Another sign is too much time to convergence (or no convergence)*
- ☛ *Most common problem: your model is overspecified*
- ☛ *Most common solution: simplify the model*

*Many practical strategies discussed in ALDA, Section 5.2.2*

*Another major advantage of the multilevel model for change:  
How easy it is to include **time-varying predictors***

# Discontinuous Individual Change

# Types of discontinuous level-1 models for change

(Singer and Willet pgs. 191-201)

- Examples of discontinuous change
  - Lifetime depression trajectories
    - Relationship status
    - Children
  - Infant development trajectories
  - Wage trajectories
    - GED
    - Certifications

# Types of discontinuous level-1 models for change

(Singer and Willet pgs. 191-201)

- Must know *why* and *when* the change occurs
- Can insert 1 or more time varying predictors in growth model to model discontinuity

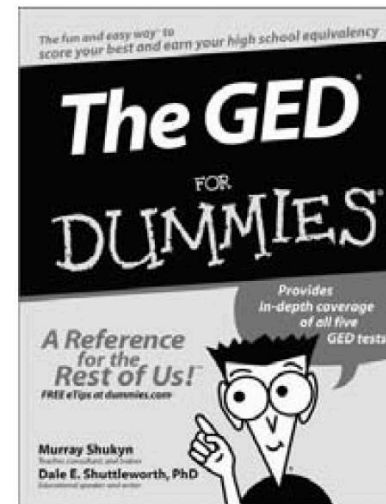
## Conceptualizing Discontinuity First:

1. Immediate shift in slope but none in elevation? (Model C)
2. Immediate shift in both elevation and slope (Model D)

# Example for discontinuous individual change: Wage trajectories & the GED

**Data source:** Murnane, Boudett and Willett (1999),  
Evaluation Review

- Sample: the same 888 male high school dropouts (from before)
- Research design
  - Each was interviewed between 1 and 13 times after dropping out
  - 34.6% (n=307) earned a GED at some point during data collection
- OLD research questions
  - How do  $\log(\text{WAGES})$  change over time?
  - Do the wage trajectories differ by ethnicity and highest grade completed?
- Additional NEW research questions: What is the effect of GED attainment? Does earning a GED:
  - affect the wage trajectory's **elevation**?
  - affect the wage trajectory's **slope**?
  - create a **discontinuity** in the wage trajectory?



The researchers hypothesized that dropouts who obtained a GED would have higher salaries than those without a GED

# Types of discontinuous level-1 models for change

(Singer and Willet pgs. 191-201)

- National Longitudinal Survey of Youth (NLSY) example
  - **Does experience influence trajectories of wages?**
  - Attempting to understand labor market experiences of male high school dropouts
  - Natural logarithm of wages (**LWN**) in years
  - The moment to the nearest day in an individuals work history (**EXPER**)
  - Highest Grade Completed 6-12 (**HGC**); Centered at 9<sup>th</sup> grade (**HGC-9**)
  - Race/Ethnicity – Black vs. White/Latino (**BLACK**)
  - Pre- or Post- General Education Development diploma (GED) at any given time point (**GED**)

The researchers hypothesized that dropouts who obtained a GED would have higher salaries than those without a GED

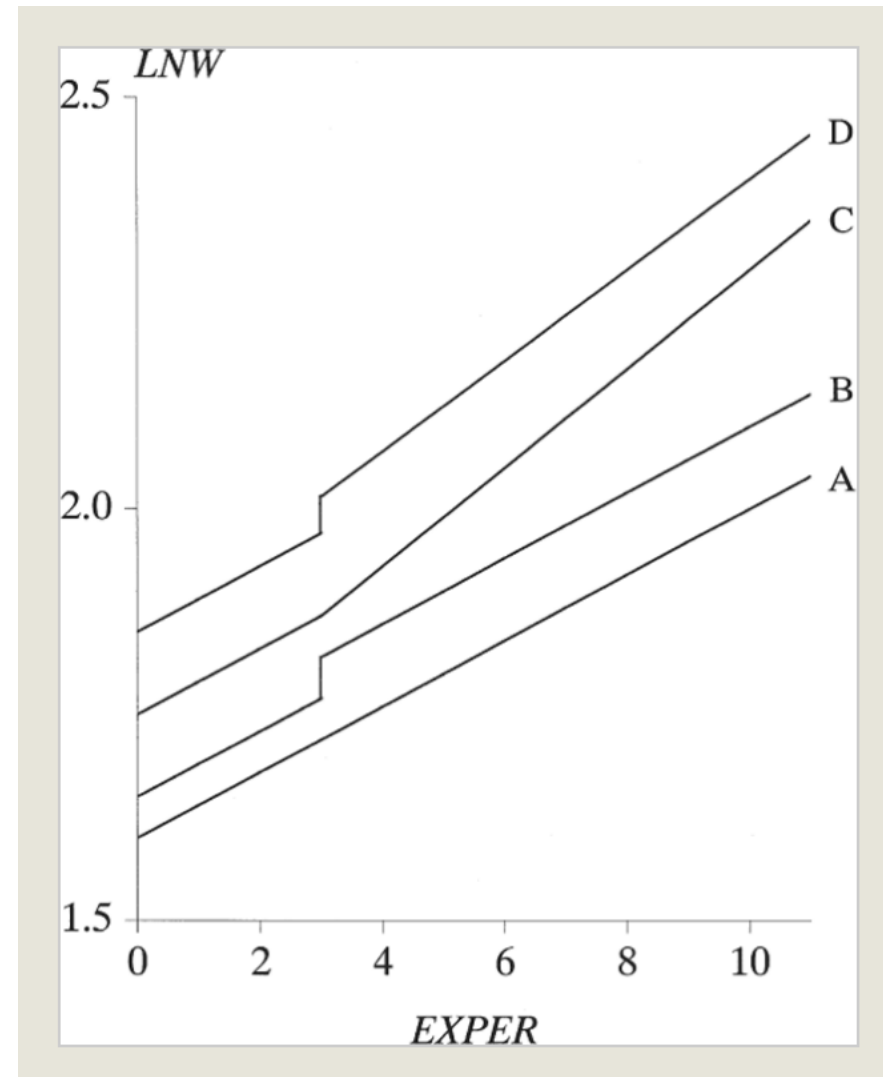
# Types of discontinuous level-1 models for change

(Singer and Willet pgs. 191-201)

$$Y_{ij} = \pi_{0i} + \pi_{1i} \text{EXPER}_{ij} + \varepsilon_{ij}$$

## Different Discontinuous Trajectories

- A** – GED does not influence trajectory (**Linear Trajectory**)
- B** – Initial increase (elevation) in wage, but not difference in rate of change in wage over time
- C** – No Initial increase in wage, but an immediate increase in the rate of change in wage over time
- D** – Initial increase in wage and immediate increase in the rate of change in wage over time



# Types of discontinuous level-1 models for change

(Singer and Willet pgs. 191-201)

## B – Initial increase in wage, but no difference in rate of change in wage over time

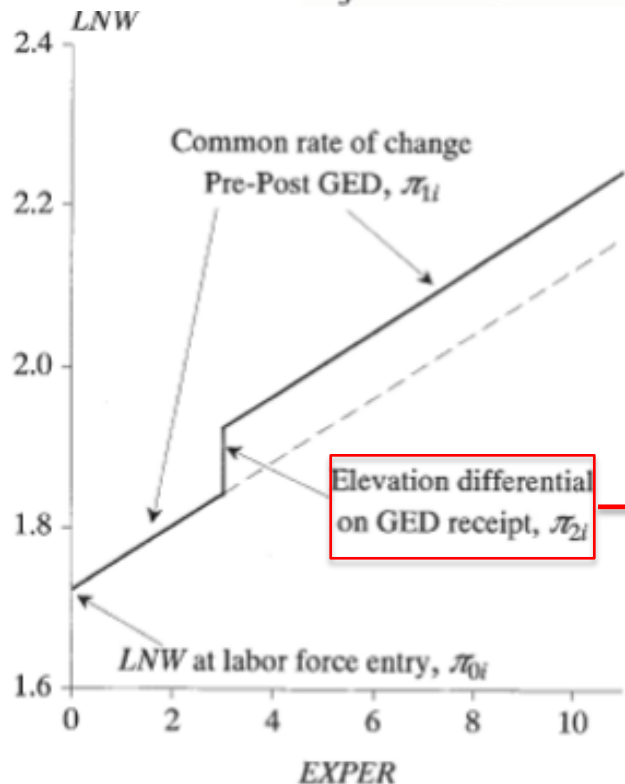
$$Y_{ij} = \pi_{0i} + \pi_{1i}EXPER_{ij} + \pi_{2i}GED_{ij} + \varepsilon_{ij}$$

Post-GED ( $GED=1$ ):

$$Y_{ij} = (\pi_{0i} + \pi_{2i}) + \pi_{1i}EXPER_{ij} + \varepsilon_{ij}$$

Pre-GED ( $GED=0$ ):

$$Y_{ij} = \pi_{0i} + \pi_{1i}EXPER_{ij} + \varepsilon_{ij}$$



Adding the time-varying covariate GED to the unconditional growth model will allow us to see initial increases in wages when/if individual “i” receives his GED

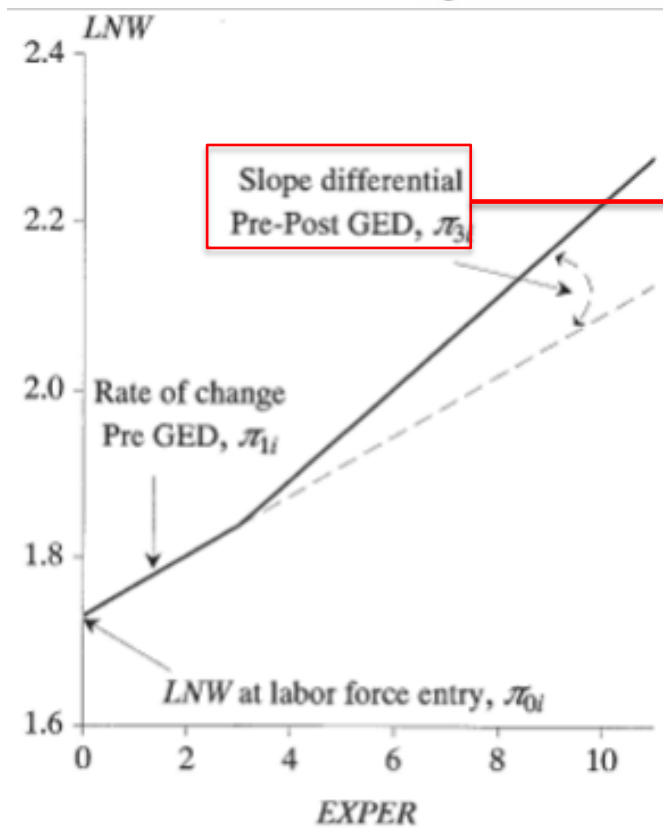
This term gives us the differential between pre and post GED

# Types of discontinuous level-1 models for change

(Singer and Willet pgs. 191-201)

C – No Initial increase in wage, but an immediate increase in the rate of change in wage

$$Y_{ij} = \pi_{0i} + \pi_{1i} \text{EXPER}_{ij} + \pi_{3i} \text{POSTEXP}_{ij} + \varepsilon_{ij}$$

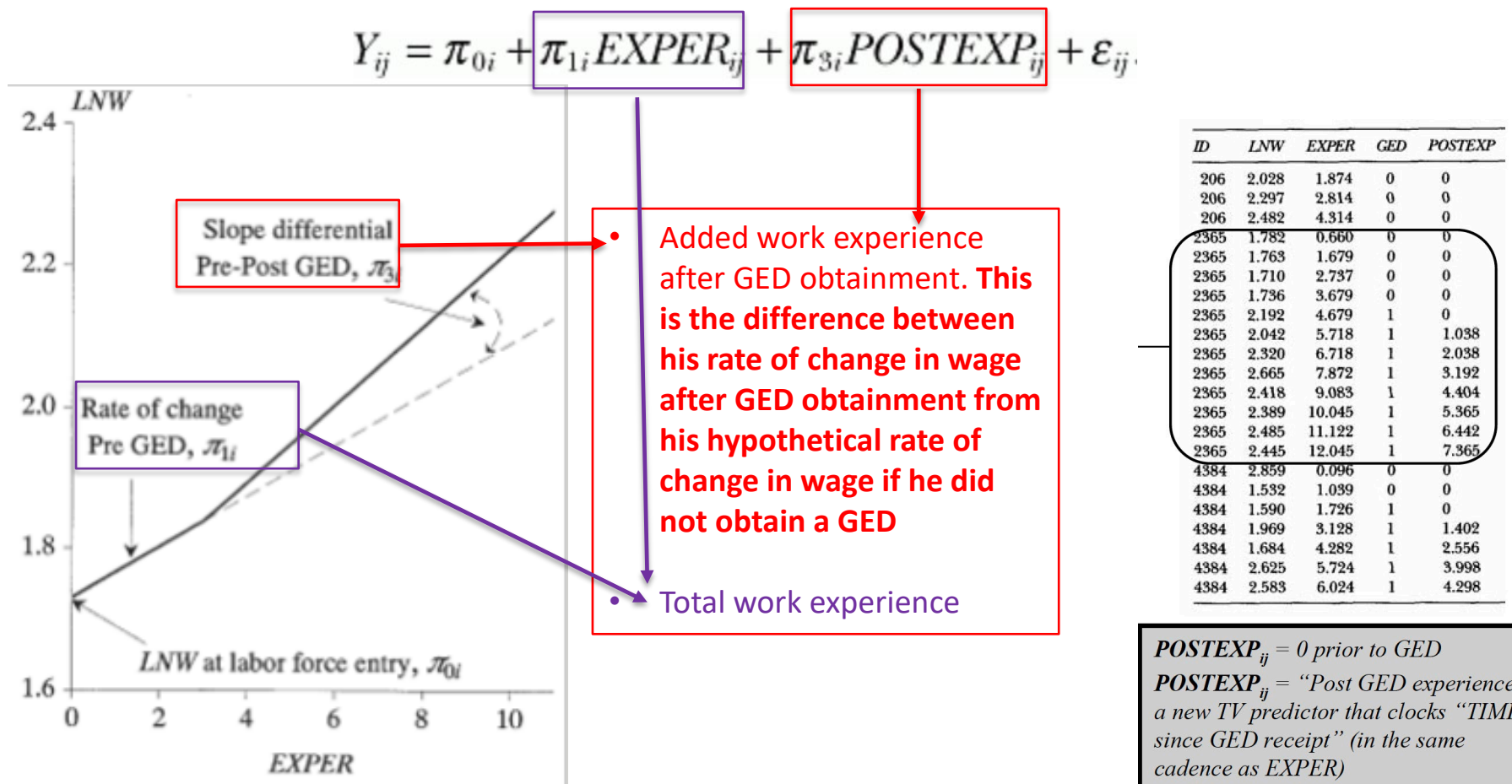


- To understand the discontinuity in the slope we must add a new variable. The rate of change between pre and post GED obtainment
- The new variable, POSTEXP, calculates labor force participation from the day the GED was obtained. Thus, up until GED obtainment participants will have a “0” value. The day after they will be assigned a “1” value

# Types of discontinuous level-1 models for change

(Singer and Willet pgs. 191-201)

C – No Initial increase in wage, but an immediate increase in the rate of change in wage



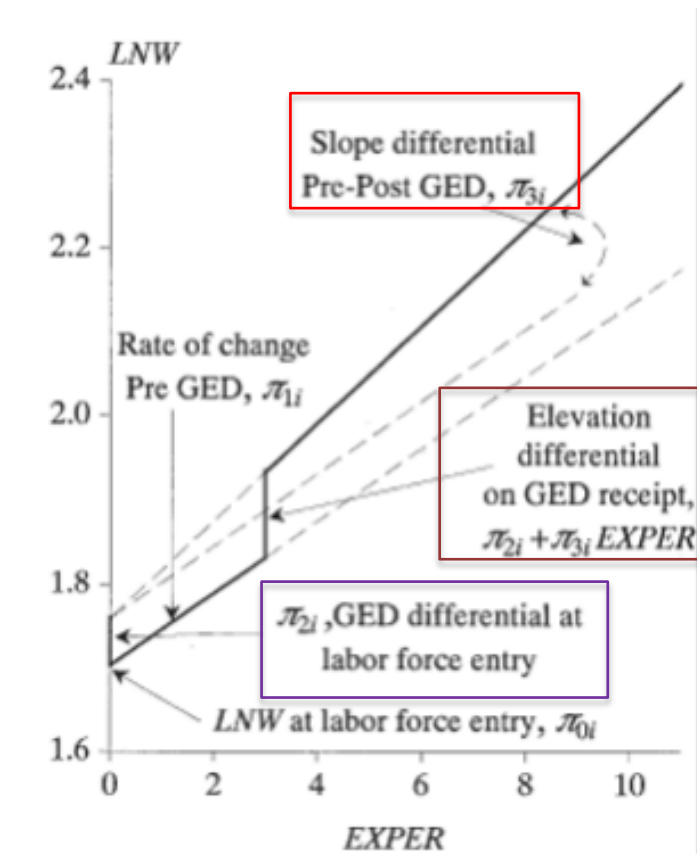
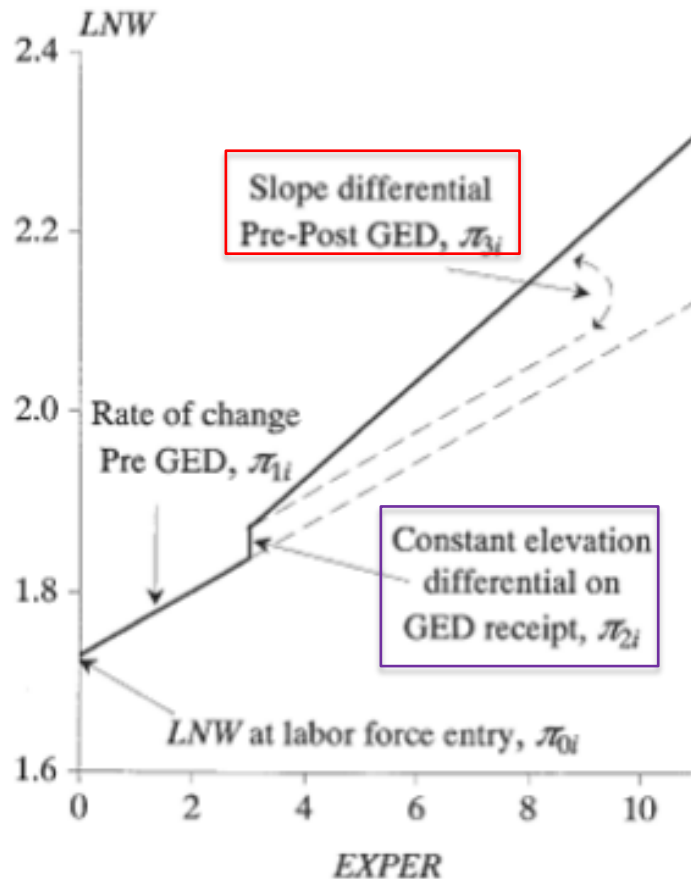
# Types of discontinuous level-1 models for change

(Singer and Willet pgs. 191-201)

## D – Initial increase in wage and immediate increase in the rate of change in wage

$$Y_{ij} = \pi_{0i} + \pi_{1i} \text{EXPER}_{ij} + \pi_{2i} \text{GED}_{ij} + \pi_{3i} \text{POSTEXP}_{ij} + \varepsilon_{ij}$$

$$Y_{ij} = \pi_{0i} + \pi_{1i} \text{EXPER}_{ij} + \pi_{2i} \text{GED}_{ij} + \pi_{3i} (\text{GED}_{ij} \times \text{EXPER}_{ij}) + \varepsilon_{ij}$$



# Selecting the best model of discontinuous level-1 change

(Singer and Willet pgs. 202-206)

**Table 6.2: Comparison of fitting alternative discontinuous change trajectories to the high school dropout wage data ( $n = 888$ )**

| Model | Fixed effects                                                                                    | Variance components<br>(in addition to $\sigma^2_{\epsilon}$ )          | n parameters for ... |                     | Deviance | Comparison model:<br>$\Delta$ Deviance (df) |
|-------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------|---------------------|----------|---------------------------------------------|
|       |                                                                                                  |                                                                         | Fixed effects        | Variance components |          |                                             |
| A     | Intercept, <i>EXPER</i> , <i>HGC</i> – 9, <i>BLACK</i> $\times$ <i>EXPER</i> , <i>UERATE</i> – 7 | Intercept, <i>EXPER</i>                                                 | 5                    | 4                   | 4830.5   | —                                           |
| B     | Model A + <i>GED</i>                                                                             | Intercept, <i>EXPER</i> , <i>GED</i>                                    | 6                    | 7                   | 4805.5   | A: 25.0*** (4)                              |
| C     | Model B                                                                                          | Model B w/o <i>GED</i>                                                  | 6                    | 4                   | 4818.3   | B: 12.8** (3)                               |
| D     | Model A + <i>POSTEXP</i>                                                                         | Intercept, <i>EXPER</i> , <i>POSTEXP</i>                                | 6                    | 7                   | 4817.4   | A: 13.1** (4)                               |
| E     | Model D                                                                                          | Model D w/o <i>POSTEXP</i>                                              | 6                    | 4                   | 4820.7   | D: 3.3 (ns) (3)                             |
| F     | Model A + <i>GED</i> and <i>POSTEXP</i>                                                          | Intercept, <i>EXPER</i> , <i>GED</i> , <i>POSTEXP</i>                   | 7                    | 11                  | 4789.4   | B: 16.2** (5)                               |
|       |                                                                                                  |                                                                         |                      |                     |          | D: 28.1*** (5)                              |
| G     | Model F                                                                                          | Model F w/o <i>POSTEXP</i>                                              | 7                    | 7                   | 4802.7   | F: 13.3** (4)                               |
| H     | Model F                                                                                          | Model F w/o <i>GED</i>                                                  | 7                    | 7                   | 4812.6   | F: 23.3*** (4)                              |
| I     | Model A + <i>GED</i> and <i>GED</i> $\times$ <i>EXPER</i>                                        | Intercept, <i>EXPER</i> , <i>GED</i> , <i>GED</i> $\times$ <i>EXPER</i> | 7                    | 11                  | 4787.0   | B: 18.5*** (5)                              |
| J     | Model I                                                                                          | Model I w/o <i>GED</i> $\times$ <i>EXPER</i>                            | 7                    | 7                   | 4804.6   | I: 17.6** (4)                               |

$\sim p < .10$ ; \*  $p < .05$ ; \*\*  $p < .01$ ; \*\*\*  $p < .001$ .

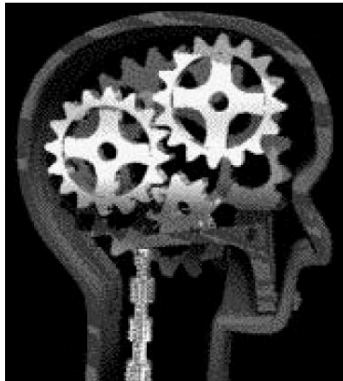
# *Many other types of discontinuous individual change trajectories are possible*

*Just like a regular regression model, the multilevel model for change can include discontinuities, non-linearities and other 'non-standard' terms*

- ◆ Generally more limited by data, theory, or both, than by the ability to specify the model
- ◆ Extra terms in the level-1 model translate into extra parameters to estimate

*What kinds of other complex trajectories could be used?*

- ◆ Effects on elevation and slope can depend upon timing of GED receipt (ALDA pp. 199-201)
- ◆ You might have non-linear changes before or after the transition point
- ◆ The effect of GED receipt might be instantaneous but not endure
- ◆ The effect of GED receipt might be delayed
- ◆ Might there be multiple transition points (e.g., on entry in college for GED recipients)



*Think carefully about what kinds of discontinuities might arise in your substantive context*

*How do we select among the alternative discontinuous models?*

# Summary

- You can model discontinuity at level 1
- Must have an apriori idea of what discontinuity looks like
  - Sketch out on a piece of paper
  - Take a look at the academic research literature
- Once you have some ideas you can attempt to model this change
- Next, you should assess model fit through running different models and comparing the deviance statistic
- Once you have modeled the different types of change and selected the best statistical model, you must go back to your apriori ideas, conceptual models, etc. to select a final model

Questions ?

# Administrative Stuff for Class Project

**Class project (one submission per team at Brightspace!!!)**

- **Midnight on Friday, March 21, 2025: Proposal** for class project due at Brightspace  
(Every submitted team receives full credits)
- **12noon on Thursday, April 24, 2024, proposal presentation file** due at Brightspace
- **Midnight on Thursday, May 8, 2025, written report** due at Brightspace

# Next Class...

**A brief introduction to using polynomials** (Singer and Willett, pgs. 213-223)

**Logistic models and Fixed effect models** (readings are posted to NYU Classes)

- Logistic regression
  - How do these models differ from linear models?
  - Interpretation of coefficients
  - Things to consider
- Fixed effect model
  - How do these models differ from linear and logistic models?
  - Interpretation of coefficients
  - Things to consider

# Class Exercise

Please complete the lecture 8 example

## **Directions:**

1. Create the OLS trajectory graph on pg. 143; Figure 5.1 (CNLSY dataset)
2. Run the unconditional growth models using the two age variables presented on pg. 145; Table 5.2
3. Run the unconditional and conditional growth models on page 149; Table 5.4
4. Create the figure on pg. 150